

Are Sustainable Development Goals delivering convergence?



Group 13

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HSC-202 Presentation Report

April, 2026

Contents

1	Motivation & Core Question	1
2	Why Convergence Matter	1
3	SDG Performance	2
4	Convergence Concepts	2
5	Theoretical Framework	3
6	Hypotheses	3
7	Data and Measurement	4
8	σ-Convergence Model	4
9	σ-Convergence Results	5
10	β-Convergence Model	6
11	β-Convergence Results	7
12	Speed of Convergence	8
13	Key Structural Insight	8
14	σ vs β Paradox	8
15	Club Convergence	9
16	Case Study 1: Nordic vs Fragile States	10
17	Case Study 2: India vs USA	10
18	Final Empirical Answer	11
19	Policy Implications	11
20	Conclusion	12
	References	13

1

Motivation & Core Question

The 2030 Agenda (2015) defines 17 SDGs as a universal development compact across social, economic, and environmental dimensions [6, 4]. The practical policy question is not only whether average SDG performance improves, but whether the distance between high-performing and low-performing countries narrows over time.

Stylized fact: cross-country inequality remains large. In recent SDSN scorecards, frontier countries are typically near 80 points while fragile states often remain far below 50 [4].

To formalize the gap, we track:

$$\Delta_t = \bar{S}_t^{\text{high}} - \bar{S}_t^{\text{low}}$$

where $\bar{S}_t^{\text{high}}$ and $\bar{S}_t^{\text{low}}$ denote average SDG scores of top and bottom country groups at time t . A persistent or rising Δ_t implies weak equalization even when global means improve.

Core research question: are countries converging in SDG outcomes, or are SDGs improving global averages without sufficiently reducing global disparities?

2

Why Convergence Matter

Convergence is central because SDGs are normatively about inclusion. If lower-performing countries systematically catch up, the framework is reducing development inequality; if not, the system may deliver progress without fairness [4].

Let cross-country welfare loss from inequality be:

$$\mathcal{L}_t = \sum_{i=1}^N (S_{it} - \bar{S}_t)^2$$

where S_{it} is the SDG composite for country i and $\bar{S}_t$ is the global mean. Convergence corresponds to a decline in $\mathcal{L}_t$ over time.

Policy meaning:

- If convergence exists, multilateral frameworks and diffusion mechanisms are working.
- If convergence is weak, place-based and institution-sensitive policies are required.

This is why convergence analysis is a stronger evaluation criterion than average growth alone [1, 5].

3

SDG Performance

The SDSN SDG Index aggregates goal-level performance into a 0–100 score [6, 4]. A stylized representation is:

$$S_{it} = 100 \sum_{g=1}^{17} w_g s_{igt}, \quad \sum_{g=1}^{17} w_g = 1$$

where s_{igt} is normalized performance for goal g in country i at time t , and w_g are aggregation weights.

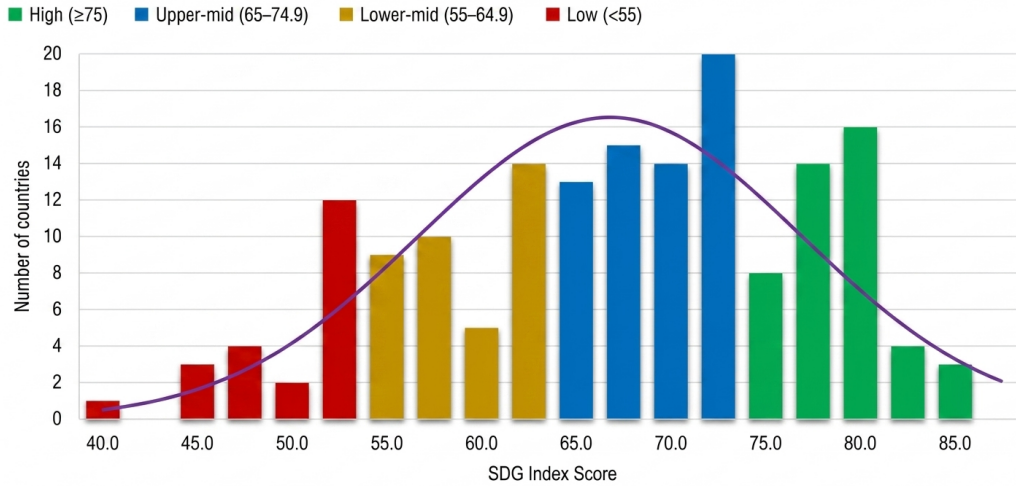


FIGURE 3.1: SDG Index Distribution Across Countries.[4]

Empirically, the global distribution is left-skewed: many countries cluster in mid-to-high ranges, but a non-trivial tail remains far behind. Even top-ranked countries exhibit persistent deficits in goals like SDG 2 (Hunger) and SDG 13 (Climate Action), so complete achievement remains elusive [4].

4

Convergence Concepts

We use four complementary concepts from the growth-convergence literature [1, 5, 3]:

- **σ -convergence:** cross-country dispersion falls over time, i.e., $\sigma_{t+1} < \sigma_t$.
- **β -convergence:** countries with lower initial SDG levels grow faster (negative slope in growth-on-initial-level regression).
- **Conditional convergence:** countries converge after controlling for structural differences such as income, institutions, and demographics.
- **Club convergence:** convergence occurs within homogeneous groups, not necessarily across the whole global sample.

In practice, robust diagnosis requires reading all four together because any single metric can be misleading in heterogeneous global panels.

5

Theoretical Framework

SDG dynamics can be represented as a balance between policy effort and structural frictions:

$$\Delta S_{it} = \alpha + \phi I_{it} - \theta B_{it} + u_{it}$$

where I_{it} captures effective investment and policy intensity, and B_{it} captures barriers such as conflict, weak institutions, and geographic constraints.

Two mechanisms jointly operate:

- **Diminishing returns and catch-up:** low-initial performers may gain more from each incremental intervention, supporting convergence.
- **Structural persistence:** institutional weakness and fragility can offset catch-up forces, sustaining divergence.

This dual mechanism is consistent with classical convergence theory and modern panel evidence in SDG applications [1, 5, 2].

6

Hypotheses

Based on theory and recent empirical findings, we test:

- **H1:** $\beta < 0$ for the aggregate SDG index and for most individual goals.
- **H2:** Global σ -convergence is weak or absent over the full period.
- **H3:** Conditional controls increase evidence of convergence.
- **H4:** Club convergence is stronger than global convergence.

These hypotheses are aligned with recent SDG convergence studies [2, 3].

7

Data and Measurement

Sample and sources.

- Country panel: roughly 190 countries, 2000–2022.
- SDG indicators: UN/SDSN repositories.
- Macro controls: World Bank WDI.

Normalization. For comparability, each indicator is transformed to a 0–100 scale:

$$s_{igt} = 100 \times \frac{x_{igt} - x_g^{\min}}{x_g^{\max} - x_g^{\min}}$$

with direction adjusted where lower raw values imply better outcomes.

Measurement strategy. We report aggregate SDG index results and goal-wise estimations. To reduce missingness bias, each goal is proxied by one primary indicator in the baseline specification, then checked for robustness with alternative indicators where feasible [2, 6].

8

σ -Convergence Model

The dispersion metric is:

$$\sigma_t = \sqrt{\frac{1}{N} \sum_{i=1}^N (S_{it} - \bar{S}_t)^2}$$

where $\bar{S}_t = \frac{1}{N} \sum_i S_{it}$.

To account for changes in the global mean, we also monitor relative dispersion:

$$CV_t = \frac{\sigma_t}{\bar{S}_t}$$

Convergence requires a declining σ_t (and preferably declining CV_t). If dispersion is flat or increasing, global inequality remains high despite average gains [5, 2].

9

σ -Convergence Results

Empirical evidence suggests weak global σ -convergence, with rising dispersion indicating persistent cross-country inequality [2]. Goal-level patterns are heterogeneous, requiring a disaggregated view. Poverty- and access-related goals (e.g., water, sanitation, energy) show clearer convergence, largely driven by technological diffusion and international aid. In contrast, structurally complex challenges—such as hunger, inequality, and climate sustainability—exhibit persistent or increasing divergence.

This contrast highlights a structural insight: global policies are more effective for discrete, infrastructure-based goals that can bypass state capacity constraints. Goals requiring institutional depth, behavioral change, or redistribution see weaker outcomes, as lower-capacity states struggle to implement them effectively. Consequently, improvements in global averages can obscure widening disparities in critical areas. These dynamics suggest that policy frameworks must shift toward context-specific institutional support, as global averages alone do not reflect equitable reductions in inequality [4, 2].

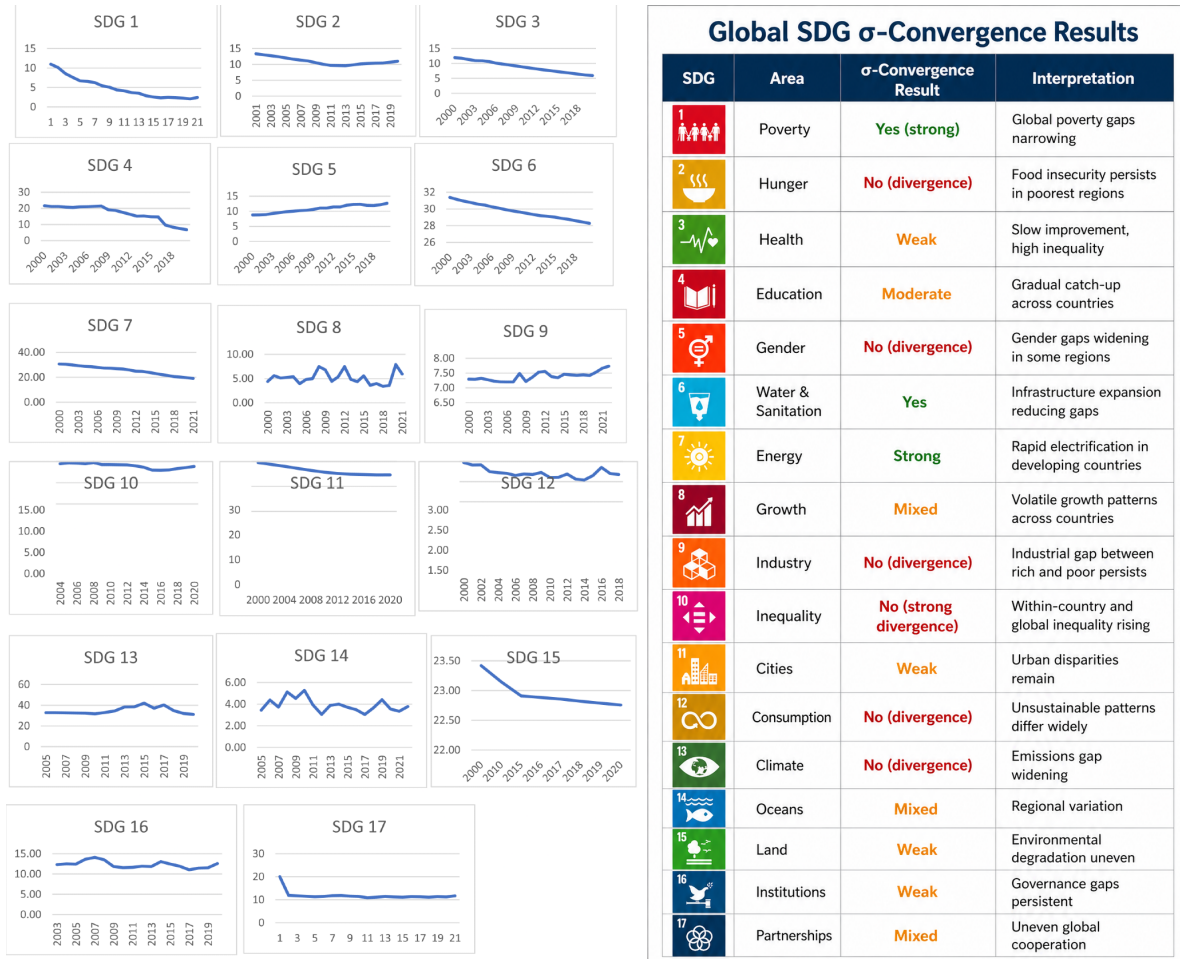


FIGURE 9.1: σ -convergence patterns across SDGs (adapted from 2).

10

β -Convergence Model

We estimate absolute and conditional β -convergence using panel regressions:

$$\Delta \ln S_{it} = \alpha + \beta \ln S_{it-1} + \gamma' X_{it} + \mu_i + \tau_t + \varepsilon_{it}$$

where X_{it} includes controls (income, institutions, demographics), μ_i are country effects, and τ_t are time effects.

Interpretation of β :

- $\beta < 0$: lower-initial countries grow faster (catch-up).

- $\beta = 0$: no systematic catch-up.
- $\beta > 0$: divergence.

Because SDG panels exhibit persistence and potential endogeneity, baseline OLS estimates are complemented with dynamic panel GMM (Arellano–Bond style) for robustness [1, 2].

11

β -Convergence Results

Most goals show statistically significant negative β estimates, indicating meaningful catch-up dynamics. Convergence is generally faster in market-linked domains (energy, infrastructure, growth) and slower in institution-intensive domains (hunger, health, inequality) [2, 4].

Table 2
Unconditional β -convergence – SDG World countries.

<i>Dep. variable: $\Delta \ln(X/X_{t-1})$</i>	SDG 1 OLS	SDG 1 GMM	SDG 2 OLS	SDG 2 GMM	SDG 3 OLS	SDG 3 GMM	SDG 4 OLS	SDG 4 GMM
<i>Indep. Var. (X_{t-1}/X_{t-2})</i>	−1.61 **	−1.21 **	−0.33 **	−0.27 **	−0.20 **	−0.11 **	−1.03 **	−1.04 **
<i>R-squared</i>	0.34	0.41	0.44	0.44	0.64	0.62	0.01	
<i>N. obs.</i>	554	268	2376	2112	3100	2790	90	90
<i>Speed of conv.</i>	0.19	0.16	0.057	0.047	0.037	0.021	0.35	0.35
<i>Hausmann p-value</i>		0.01		0.01		0.01		0.41
<i>Half-life</i>	3.59	4.34	12.2	14.5	18.71	32.08	2.0	1.9
<i>Dep. variable: $\Delta \ln(X/X_{t-1})$</i>	SDG 5 OLS	SDG 5 GMM	SDG 6 OLS	SDG 6 GMM	SDG 7 OLS	SDG 7 GMM	SDG 8 OLS	SDG 8 GMM
<i>Indep. variable: (X_{t-1}/X_{t-2})</i>	−0.85 **	−0.86 **	−0.86 **	−0.81 **	−0.37 **	−0.37 **	−1.53 **	−1.49 **
<i>R-squared</i>	0.01	0.02	0.02	0.04	0.39	0.40	0.24	0.24
<i>N. obs.</i>	1235	359	1558	654	1775	1775	4330	4330
<i>Hausmann p-value</i>		0.91		0.97		.73		0.07
<i>Speed of conv.</i>	0.32	0.31	0.29	0.29	0.16	0.16	0.45	0.46
<i>Half-life</i>	2.22	2.21	2.33	2.33	4.40	4.40	1.49	1.52
<i>Dep. variable: $\Delta \ln(X/X_{t-1})$</i>	SDG 9 OLS	SDG 9 GMM	SDG 10 OLS	SDG 10 GMM	SDG 11 OLS	SDG 11 GMM	SDG 12 OLS	SDG 12 GMM
<i>Indep. variable: (X_{t-1}/X_{t-2})</i>	−0.97 **	−0.97 **	−0.66 **	−0.67 **	−0.44 **	−0.45 **	−0.51 **	−0.52 **
<i>R-squared</i>	0.01	0.01	0.1	0.01	0.2	.33	.24	.24
<i>N. obs.</i>	4522	4522	651	651	60	60	502	501
<i>Hausmann p-value</i>		0.92		0.59		.98		.77
<i>Speed of conv.</i>	0.33	0.34	0.25	0.26	0.18	0.18	0.21	0.21
<i>Half-life</i>	2.04	2.05	2.70	2.71	3.74	3.73	3.33	3.34
<i>Dep. variable: $\Delta \ln(X/X_{t-1})$</i>	SDG 13 OLS	SDG 13 GMM	SDG 14 OLS	SDG 14 GMM	SDG 15 OLS	SDG 15 GMM	SDG 16 OLS	SDG 16 GMM
<i>Indep. variable: (X_{t-1}/X_{t-2})</i>	−0.47 **	−0.47 **	−0.78	−0.77 **	−0.37 **	−0.39 **	−0.82 **	−0.82
<i>R-squared</i>	0.01	0.25	0.05	0.05	0.57	0.58	0.03	0.03
<i>N. obs.</i>	270	270	359	359	286	286	373	373
<i>Hausmann p-value</i>		0.88		0.16		0.03		0.79
<i>Speed of conv.</i>	0.19	0.19	0.27	0.28	0.15	0.16	0.30	0.30
<i>Half-life</i>	3.60	3.60	2.39	2.42	4.40	4.22	2.31	2.32
<i>Dep. variable: $\Delta \ln(X/X_{t-1})$</i>	SDG 17 OLS	SDG 17 GMM						
<i>Indep. variable: (X_{t-1}/X_{t-2})</i>	−0.83 **	−0.84 **						
<i>R-squared</i>	0.03	0.03						
<i>N. obs.</i>	270	764						
<i>Hausmann p-value</i>		0.11						
<i>Speed of conv.</i>	0.30	0.30						
<i>Half-life</i>	2.28	2.28						

Notes: (i) sample period: generally 2000 – 2022, availability may change for some indicators; (ii) ** denotes significance at 1 % confidence level; * at 5 % level.

FIGURE 11.1: Estimated unconditional β -convergence coefficients and implied speeds (compiled from reported estimates in 2).

The key empirical message is that β -convergence can coexist with weak σ -convergence: many countries improve faster, yet overall inequality can remain high when initial gaps are very large.

12

Speed of Convergence

With β estimated from the growth specification, the annual convergence speed can be approximated by:

$$\lambda = -\ln(1 + \beta), \quad (\beta > -1)$$

and the half-life of the gap is:

$$T_{1/2} = \frac{\ln 2}{\lambda}$$

If λ is high, catch-up is rapid; if λ is low, progress is slow even when convergence exists. Reported ranges in the SDG slide evidence imply half-lives that vary from a few years to multiple decades, highlighting strong heterogeneity across goals [2].

13

Key Structural Insight

A useful organizing distinction is between market-oriented goals and policy/institution-oriented goals.

1. **Market-oriented SDGs:** often converge faster because technology diffusion and private investment scale quickly (for example energy access and infrastructure).
2. **Policy/institution-oriented SDGs:** often converge slower because they require deep administrative capacity, social trust, and sustained public-system quality.

This explains why average global progress can coexist with persistent gaps in institution-heavy goals [4].



Figure 1. SDGs market-oriented and policy-oriented.
Source: our elaboration

FIGURE 13.1: Illustrative classification of SDGs by implementation channel
(image from project material; discussion linked to 4, 2).

14

σ vs β Paradox

The paradox is resolved by decomposition of total inequality into within-group and between-group components:

$$\text{Var}(S_t) = \sum_c p_c \text{Var}(S_t | c) + \text{Var}(\mathbb{E}[S_t | c])$$

where c indexes country clubs and p_c is club population share.

Even when many countries catch up within clubs (negative β), between-club gaps can stay large, keeping total dispersion high. Hence, β -convergence and weak global σ -convergence are not contradictory; they describe different margins of adjustment [3, 2].

15

Club Convergence

Following Phillips–Sul style intuition, define transition paths relative to the global mean:

$$h_{it} = \frac{S_{it}}{\frac{1}{N} \sum_{j=1}^N S_{jt}}$$

Convergence requires cross-sectional variation in h_{it} to shrink over time. In practice, evidence suggests that countries with similar initial conditions converge internally, while distances between high- and low-performing clubs remain substantial [3, 2].

This club perspective is policy-relevant: one global target architecture requires differentiated national pathways.

16

Case Study 1: Nordic vs Fragile States

Nordic countries (for example Finland and Sweden) are near the SDG frontier because of strong institutions, administrative quality, and durable social investment. Fragile states (for example Yemen and Somalia) remain in low-score equilibria due to conflict, fiscal stress, and weak public-service capacity [4].

The divergence can be interpreted as a structural-parameter gap in the framework from Chapter 5:

$$\Delta S = \phi I - \theta B$$

with lower effective ϕ and higher θ in fragile contexts. Therefore, identical global frameworks produce very different realized trajectories.

17

Case Study 2: India vs USA

India illustrates high marginal gains from targeted social and infrastructure policies (poverty reduction and electrification), while the United States illustrates a high baseline with persistent challenges in inequality and emissions intensity [6, 4].

This comparison highlights multidimensional trade-offs: advances in one domain do not automatically translate into progress in all domains. A simple representation is:

$$\max U = f(S^{\text{social}}, S^{\text{economic}}, S^{\text{environment}})$$

subject to fiscal, institutional, and political constraints. Countries differ in constraints and therefore follow distinct feasible SDG paths.

18

Final Empirical Answer

The combined evidence supports a balanced conclusion:

- **β -convergence:** broadly present across many goals.
- **σ -convergence:** partial and weak at global aggregate level.
- **Club convergence:** strong and policy-relevant.

Hypothesis assessment is therefore:

- H1 supported for most goals,
- H2 supported at aggregate level,
- H3 supported when controls are included,
- H4 strongly supported by heterogeneous trajectories.

These findings are consistent with both classical convergence theory and current SDG evidence [1, 5, 2, 3].

19

Policy Implications

Evidence implies that “one framework” cannot mean “one policy design”.

- **Target lagging countries:** prioritize fragile and low-capacity states with concessional finance and technical implementation support.
- **Prioritize slow-converging goals:** hunger, health, and inequality need long-horizon institutional investment.
- **Strengthen state capacity:** governance quality is a first-order determinant of sustainable catch-up.
- **Leverage private capital where scalable:** energy, connectivity, and infrastructure can converge faster when regulatory design is credible.
- **Track dispersion, not only averages:** monitoring frameworks should include σ_t , club gaps, and goal-specific half-lives.

These implications align with recent SDG financing and implementation priorities [4, 6].

20

Conclusion

Global SDG performance has improved, but equalization remains incomplete. The empirical structure is best summarized as “progress with stratification”: widespread catch-up in several domains, weak reduction in aggregate inequality, and strong club dynamics.

Therefore, the central policy lesson is differentiation. Countries require tailored strategies based on initial conditions, institutional capacity, and goal-specific bottlenecks. SDGs can raise development levels globally, but reducing cross-country inequality requires explicit convergence-oriented design and sustained institutional deepening [4, 2, 3].

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